Stabilization of a 3D-Printed UAV Using Consumer-Grade IMU Feedback

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Abstract

Within this project we will investigate the merit of utilizing a low-cost inertial measurement unit (IMU) and a Raspberry Pi Pico microcontroller to stabilize a fully 3D printed quadcopter UAV. A baseline will be established by recording natural drift and vibrations within the system. Then it will be contrasted with the data acquired when a proportional-integral-derivative (PID) controller, supplied by data from a MPU-6056, is active. Low cost sensors are accompanied by sensor noise and bias, to reduce that, complementary and Kalman methods are being considered. Improvements to stability directly related to the usage of consumer grade sensors, are the results that are being targeted. Overall this work aims to increase accessibility of low cost UAVs for educational purposes, such as laboratory research and STEM outreach purposes.

1 Introduction

Unmanned aerial vehicles (UAVs)are rapidly showcasing their merit across various aspects of society. Ranging from research, to commercial and to educational domains. Their versatility and modularity makes them a prime choice for varying applications such as environmental monitoring, wireless communication networks, delivery systems and advanced vertical robotics education [1, 2]. However despite their large scale availability, they are limited due to software and hardware compatibility issues imposed by their makers. Resulting in various makers and hobbyist commutes creating do-it-yourself (DIY) builds, from consumer grade and 3D printed components. Demonstrating that low cost UAVs can be created [3]. Recent efforts have been made in professional engineering circles to simplify and accelerate drone control, to the point in which quadcopters can be stabilized within mere moments [4].

Stability is one of the largest challenges to designing a UAV. Older models of UAV frames used in commercial drones are structural sound and resist vibration, however they are costly and less reproducible. However 3D printed UAV frames, despite being more affordable and modular, have a higher resonance and structural vibration [5, 6, 7, 8]. Additionally low cost IMU's do offer stability enhancements, however they are limited by reliability, being plagued by drift, noise and bias [9, 10, 11]. Research suggests that various data filtering and control techniques will minimize these limitations, however they are seldom utilized simultaneously with low cost 3D printed UAV frames.

By investigating a 3D printed quadcopter UAV paired with a consumer grade IMU and controlled by a Raspberry Pi Pico, this project aims to bridge the gap between low cost and well funded UAV projects. Allowing for the measurement of improvements directly linked to low cost stability enhancements as a result of sensors and fabrication methods. By comparing baseline drift to its counterpart under the effects of a PID controller. Summarizing the intent of this project is to showcase that low cost UAV designs can provide meaningful experience and date, helping to extend teaching opportunities and independent research in this field.

2 Design and Methods

The typical UAV has several key components. Those being the 3D-printed frame, brushless motors with propellers, a microcontroller, and an inertial measurement unit (IMU). Each part is a vital component to overall flight, stabilization and with proper integration they will all compound in increasing stability.

2.1 Frame and Materials

The airframe has to be fabricated using fused deposition modeling (FDM), with PLA being selected for its ease of printing. ABS and PETG were also strong candidates due to their toughness and larger vibration resistance. Based on studies, infill density, orientation and layer height have a direct correlation to stiffness and vibration damping [6, 7, 8, 12]. Simulations in conjunction with generative design demonstrate structural integrity despite significant weight reduction [5].

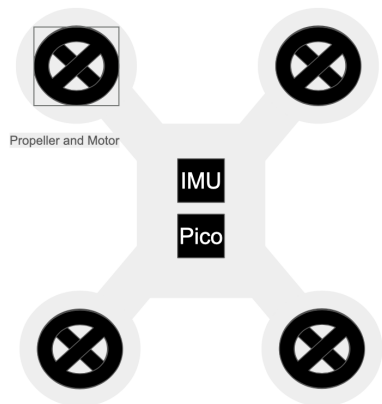

Figure 1: Representative diagram of a quadcopter UAV frame with labeled components (IMU, Pico, propellers, motors).

2.2 Motors and Propellers

Electronic speed controllers paired with four brushless motors are responsible for providing lift to the UAV. The gorla is to achieve a ratio of 2:1 (thrust-to-weight) to ensure stability during hovering. The tradeoffs between battery and motor performance have been analyzed in prior UAV studies [13], which provide a guide for choosing the best sizing for endurance and stability.

2.3 Microcontroller

A Raspberry Pi Pico was chosen to be the flight controller as it is compact, it has enough processing power to implement a PID loop and is low cost. Despite being commonly used [14], the Pico was chosen as it has a faster I/O handling and allows for both Python and iterations of C. Which aligns with the overarching trend of utilizing light weight, and low cost embedded systems for the UAV hardware [15].

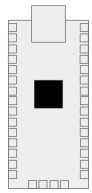

Figure 2: Raspberry Pi Pico microcontroller.

2.4 Inertial Measurement Unit (IMU)

On the other hand, the Raspberry Pi Pico micro-controller provides the perfect amount of computational power, despite its low cost. When combined with the MPU-6050, it provides the user with a simple yet effective platform to study the fundamentals of stabilizing a quadcopter. However there are still limits to it as the processing power is limited, there is sensor noise, drift and mechanical imperfections in the 3D printed frame.

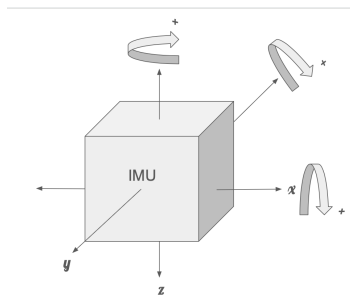

Figure 3: MPU-6050 IMU module used for feedback stabilization (3-axis accelerometer and gyroscope).

2.5 System Integration

The components need to be connected through a closed-loop system, where the IMU feeds measurements to the Pico microcontroller, which then computes the necessary corrections using the PID controller. The corrections that are generated will adjust the ESC signals that drive the motors. A massive challenge in UAV systems is having a real-time integration of sensor feedback, as any delays or noise can destabilize the UAV's flight [2]. Which is showcased in the diagram below.

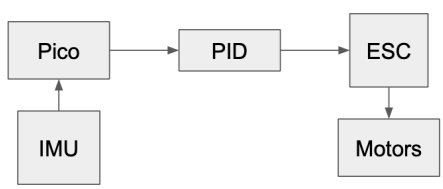

Figure 4: Control loop using IMU feedback and PID stabilization: IMU → Pico (processing) → PID → ESCs → motors.

3 Control System and Mathematical Model

Implemented by the Raspberry Pi Pico, the stability of the UAV is sustained through the proportional–integral–derivative (PID) feedback loop. The standard form of the PID control law

is

$$u(t) = K_p\, e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d\, \frac{de(t)}{dt}, \tag{1}$$

where $u(t)$ is the control signal sent to the motors, $e(t)$ is the orientation error, and τ is the integration variable of time. K_p, K_i, and K_d are the proportional, integral, and derivative gains, respectively.

Within this project, the error $e(t)$ is the difference between the orientation measured with the IMU and the desired orientation:

$$e(t) = \theta_{\text{desired}}(t) - \theta_{\text{measured}}(t), \tag{2}$$

where θ represents roll, pitch, or yaw. The microcontroller independently computes the required correction for each axis, which is then combined to adjust the speeds of the four brushless motors. The control update may be written as

$$\omega_i = \omega_{\text{throttle}} + u(t), \tag{3}$$

with ω_{throttle} the baseline motor speed for hover. Tuning balances overshoot/oscillation (too-large K_p), steady-state drift (too-small K_i), and noise amplification (too-large K_d). To increase reliability, filtering mitigates IMU noise and bias. A Kalman filter is optimal but computationally heavier; a complementary filter is lightweight [10, 11]. Stability improvements can be quantified by the reduction in angular variance:

$$\Delta\sigma^2 = \sigma^2_{\text{baseline}} - \sigma^2_{\text{PID}}. \tag{4}$$

4 Experimental Testing

Indoor tests under controlled conditions remove wind and GPS needs, reducing load on the microcontroller and IMU. Trials consist of 30–60 s hover segments with IMU logging at 200 Hz. Metrics include angular variance in roll/pitch/yaw and drift rate; frequency-domain analysis quantifies vibration magnitudes and dominant resonances. Baseline data will be compared to PID-stabilized flights. Safety: altitude < 2 m, cleared lab, spares accessible.

Broader Impacts

Overall it was concluded that UAV stabilization can be achieved utilizing low cost materials and consumer grade sensors. Lowering the entry barrier for independent research, STEM outreach programs, robotics and enabling a cheaper teaching alternative. The usage of chapter materials allows for students to gain hands-on experience with feedback systems and flight dynamics without the looming budgetary restrictions.

Additionally, it contributes to perpetual development of UAV stabilization methods, through open sourced framework, and fosters collaboration between various institutions. Directly supporting the growth of young minds, and researchers who share the intent to pursue physics or engineering, by providing them with practical exposure to modern principles and techniques.

5 Conclusion

PID stabilization is expected to produce a measurable reduction in oscillation and drift compared to free flight tests. Which would confirm that low cost IMUs and 3D printed frames can produce meaningful stability gains. Looking past the technical aspect, this work would allow for accessible low cost methods for activities such as physics and engineering education. Low cost UAV platforms would considerably enlarge the range of institutions that allow efforts to allow drove technology

within the class room [16]. Any future actions that may be taken include, generative optimization, testing alternative IMU sensors or using more adaptive control techniques.

There are a few limitations to this project. Such as it being limited to indoor testing, to avoid winds disturbing stability reading, however this restricts real world applicability. Additionally due to the nature of the consumer grade IMU, it introduces drift and noise that can't be fully removed. Lastly as a result of 3D printing the frame, it is more prone to vibration during use. These limitations showcase that there is still room for further work in the future, that includes, importing the filter, reducing vibrations, utilizing alternative materials and testing outdoors.

6 Budget

Item	**Quantity**	**Unit Cost (USD)**	**Total (USD)**
3D Printing Filament (PLA, PETG, ABS)	2–3 spools	$20	$40–60
Brushless Motors	4 + 2 spares	$8	$32–48
Propellers	3 sets (12 total)	$10	$30
Raspberry Pi Pico (microcontroller)	1	$8	$8
IMU Sensor (MPU-6050 or equivalent)	1–2	$10	$10–20
Wires, Connectors, Fasteners	Assorted	–	$15–25
Misc. Crash Replacements / Spares	–	–	$20–30
3D Printer (entry-level FDM, e.g., Ender 3 V2)	1	$200	$200
Total Estimated Budget			**$360–421**

Table 1: Estimated budget for the UAV stability project.

Budget Justification

- **3D Printing Filament (PLA, PETG, ABS):** Required for fabricating the UAV frame. PLA is chosen for ease of use.
- **Brushless Motors:** Four are required for lift, with two spares to account for mishaps.
- **Propellers:** Multiple sets are necessary because propellers are prone to damage during testing.
- **Raspberry Pi Pico (Microcontroller):** Serves as the flight controller; provides ample processing power for its size.
- **IMU Sensor (MPU-6050):** Core sensor for measuring orientation (roll, pitch, yaw).
- **Wires, Connectors, and Fasteners:** Necessary to integrate electrical components within the UAV.
- **Miscellaneous Crash Replacements / Spares:** Accounts for expendables (screws, vibration dampers, replacement mounts) expected in repeated trials.
- **3D Printer (Ender 3 V2 or equivalent):** One-time cost for producing frames and modifications.

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